THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In 1 second, the direction of a 50 Hz AC reverses 1

(a) 25 times **(b)** 200 times **(c)** 50 times **(d)** 100 times

2. A magnetic field line at a point is a curve whose tangent gives 1

(a) the direction of the current **(b)** the speed of the charge
(c) the direction of the field **(d)** the voltage

3. The force on a conductor in a magnetic field increases when 1

(a) the conductor is made parallel **(b)** the current is increased
(c) the field is reduced **(d)** the conductor is shortened to zero

4. The direction of the current in the mains supply in India reverses every 1

(a) 1/50 s **(b)** 1 s **(c)** 100 s **(d)** 1/100 s

5. A magnetic field line at a point is a curve such that its tangent shows 1

(a) the force on iron
(b) the direction of the current
(c) the direction of the field
(d) the direction of force on a charge always

6. Force on a current-carrying wire is an example of 1

(a) gravitational effect **(b)** the interaction of two magnetic fields
(c) heating effect **(d)** chemical effect

7. The device used to safely carry the leakage current to the earth is the 1

(a) earth wire **(b)** fuse **(c)** live wire **(d)** neutral wire

8. The magnetic field is strongest where 1

(a) lines cross **(b)** field lines are closest
(c) lines are straight **(d)** field lines are far

9. The SI unit of magnetic field is 1

(a) ampere metre **(b)** tesla **(c)** newton **(d)** weber per ampere

10. For a 15 A circuit, the typical appliance used is 1

(a) a phone charger **(b)** a tube light
(c) a bulb **(d)** an air conditioner or geyser

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the difference between the potential of the live wire and that of the neutral wire? 2

12. Give the reason why soft iron and not steel is used in an electromagnet. 2

13. Calculate the time period of an AC with frequency 50 Hz. 2

14. Why must the switch be on the live wire of an appliance? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. An electron enters a region of magnetic field directed into the page, moving to the right. What is the direction of the force on it? Give the reasoning. 3

16. Why does a person get a shock on touching the metal body of a faulty iron without earthing? 3

17. Why should a 2 kW heater be used with a 15 A socket, not a 5 A socket? 3

OR

Explain the working of an electric bell using an electromagnet in brief. 3

18. How would you identify the poles of a bar magnet using a compass? 3

OR

What is the effect on the magnetic field of a solenoid when an iron rod is inserted? Explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A coil of insulated wire connected to a galvanometer is kept near a bar magnet. Describe four different ways in which a current can be induced. Which factors increase the current? 5

OR

Using the idea of electromagnetic induction, explain how mechanical energy is converted to electrical energy in a power station, and the role of the coil, magnet and turbine. 5

20. Describe the working of an AC generator with the help of the position of the coil during one rotation and the direction of the current. 5

OR

Compare Oersted's experiment and Faraday's experiment in terms of the observation, the conclusion and the type of effect. 5

END OF QUESTION PAPER — Total: 40 Marks

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